

otherwise perfect crystals displaced by $a/2\langle\bar{1}01\rangle$, which is no longer a lattice vector; the associated APB energy is strongly anisotropic, being highest on the $\{110\}$ plane and lowest on $\{001\}$. Intrinsic, extrinsic and complex stacking faults have also been identified. The energies of these planar defects control the mechanism of dislocation dissociation in the γ' phase.

Since a $a/2\langle\bar{1}01\rangle\{111\}$ dislocation travelling in γ cannot enter the γ' phase without an APB forming, cutting of the γ/γ' structure requires pairs of them; this is the origin of a substantial precipitation-hardening effect known as order strengthening, which is relevant to static, tensile properties. A pair of dislocations (each one of which is known as a superpartial) may be either weakly or strongly coupled, depending upon whether their spacing is large or small in comparison with the γ' particle diameter. Cutting by the weak-coupling mechanism is applicable when the γ' particles are in the under-aged condition, but the strong-coupling one is appropriate for over-aged structures. Analysis indicates that the degree of hardening is approximately proportional to the square root of the particle radius, r , for the weak-coupling mechanism and inversely proportional to r for strong coupling; hardening is maximised for a γ' particle size at the transition between the two. For γ/γ' alloys, the yield stress does not decrease strongly as the temperature increases, provided that one stays below about 700 °C; this arises because of the anomalous yielding effect displayed by the γ' phase. The cause of this is the cross-slip of segments of the γ' superpartials from $\{111\}$ to $\{001\}$, promoted by the applied stress, the anisotropy of the APB energy and a further contribution arising from elastic anisotropy.

The creep behaviour displayed by nickel and its alloys depends strongly upon the metallurgical structure, i.e. the grain size, γ' , particle size and the concentrations of alloying additions. For polycrystalline nickel, a fifth-power-law dependence of the applied stress for the steady-state 'secondary' regime is found in common with other pure metals; this can be rationalised by balancing dislocation annihilation and multiplication processes (consistent with the 'recovery-controlled' model of creep deformation) provided that the sub-structure is assumed to coarsen with the cube root of time. Alloying additions have a strong strengthening effect on the creep of polycrystalline nickel, with strengthening being most potent when the atomic size misfit of solute and solvent is large; a distinction is made between 'Class I' behaviour, leading to a linear viscous motion of dislocations and a power-law stress dependence of three, and 'Class II' behaviour, with a dependence of five, which is analogous to the pure metals. Analysis indicates that most important alloying additions such as Fe, Co, Cr, W, Mo, Re and Ru promote Class II behaviour. The γ' phase fraction is found to have a profound influence on creep life, with the potency being greatest when the grain size is large. This effect occurs because the prominent form of creep damage in polycrystalline nickel alloys is creep cavitation at γ -grain boundaries, which becomes less influential as the grain size increases. This effect is exploited in the single-crystal superalloys, which are the topic of the Chapter 3.

Appendix. The anisotropic elasticity displayed by nickel

Nickel is *elastically anisotropic* when in single-crystal form. Hence the stiffness – a measure of the stress necessary to maintain a given strain – is dependent upon the crystallographic

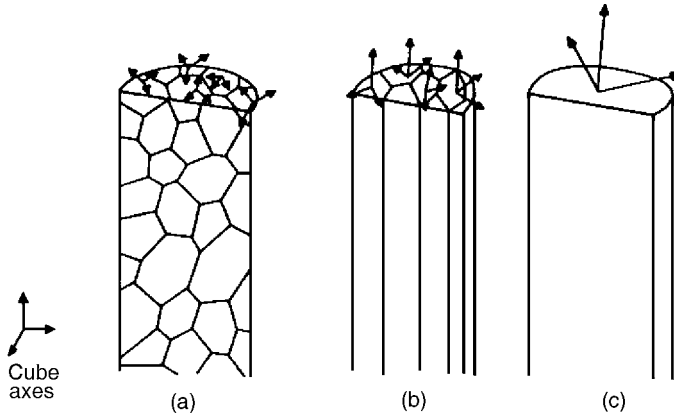


Fig. 2.74. Schematic illustration of the equiaxed, columnar and single-crystal structures used for turbine blade aerofoils.

orientation relative to the loading configuration. Consequently, specimens machined at random from a large single crystal will display different elastic properties. As turbine blading is often used in *single-crystal* or *columnar-grained* ($\langle 100 \rangle$ -textured) form, see Figure 2.74, it is important to consider this effect. Moreover, the elastic anisotropy influences the cross-slip of dislocations, particularly in γ' .

A treatment of the elastic anisotropy of nickel requires an acknowledgement that the quantity relating the stress, σ , and strain, ϵ , both second-rank tensors, will then be a fourth-rank tensor [91]. Consider a cubic single crystal with orthogonal axes 1, 2, 3 lying along the $[100]$, $[010]$ and $[001]$ directions, respectively. The elastic properties are then completely defined by *three* elastic stiffness constants C_{11} , C_{12} and C_{44} , such that

$$\begin{aligned}
 \sigma_{11} &= C_{11}\epsilon_{11} + C_{12}\epsilon_{22} + C_{12}\epsilon_{33} \\
 \sigma_{22} &= C_{12}\epsilon_{11} + C_{11}\epsilon_{22} + C_{12}\epsilon_{33} \\
 \sigma_{33} &= C_{12}\epsilon_{11} + C_{12}\epsilon_{22} + C_{11}\epsilon_{33} \\
 \sigma_{23} &= C_{44}(\epsilon_{23} + \epsilon_{32}) = 2C_{44}\epsilon_{23} \\
 \sigma_{31} &= C_{44}(\epsilon_{31} + \epsilon_{13}) = 2C_{44}\epsilon_{31} \\
 \sigma_{12} &= C_{44}(\epsilon_{12} + \epsilon_{21}) = 2C_{44}\epsilon_{12}
 \end{aligned} \tag{A1}$$

where use has been made of the reduced notation of Nye [91]. Note that the cubic symmetry reduces the number of constants in C_{ij} to three, from the 21 required for a triclinic crystal. Then $C_{11} = C_{22} = C_{33}$, $C_{44} = C_{55} = C_{66}$ and $C_{12} = C_{13} = C_{23}$; furthermore, the terms C_{14} , C_{15} , C_{16} , C_{24} , C_{25} , C_{26} , C_{34} , C_{35} , C_{36} , C_{45} , C_{46} , C_{56} are equal to zero.

An alternative representation is via the compliance constants S_{11} , S_{12} and S_{44} , such that

$$\begin{aligned}
 \epsilon_{11} &= S_{11}\sigma_{11} + S_{12}\sigma_{22} + S_{12}\sigma_{33} \\
 \epsilon_{22} &= S_{12}\sigma_{11} + S_{11}\sigma_{22} + S_{12}\sigma_{33} \\
 \epsilon_{33} &= S_{12}\sigma_{11} + S_{12}\sigma_{22} + S_{11}\sigma_{33}
 \end{aligned}$$

$$\begin{aligned}
\epsilon_{23} &= \frac{1}{2} S_{44} \sigma_{23} \\
\epsilon_{31} &= \frac{1}{2} S_{44} \sigma_{31} \\
\epsilon_{12} &= \frac{1}{2} S_{44} \sigma_{12}
\end{aligned} \tag{A2}$$

The stiffness and compliance constants are inter-related. For example, the compliance constants can be evaluated via

$$S_{11} = \frac{C_{11} + C_{12}}{(C_{11} - C_{12})(C_{11} + 2C_{12})} \quad S_{12} = \frac{C_{12}}{(C_{11} - C_{12})(C_{11} + 2C_{12})} \quad S_{44} = 1/C_{44} \tag{A3}$$

It is instructive to consider the magnitude of the constants, which have been determined by careful measurements on nickel single crystals. For pure nickel at room temperature, typical values are $S_{11} = 0.799 \times 10^{-5}/\text{MPa}$, $S_{12} = -0.312 \times 10^{-5}/\text{MPa}$ and $S_{44} = 0.844 \times 10^{-5}/\text{MPa}$. The stiffness along the $\langle 100 \rangle$ crystallographic direction is then derived by setting $\sigma_{22} = \sigma_{33} = 0$, so that

$$E_{\langle 100 \rangle} = \frac{\sigma_{11}}{\epsilon_{11}} = \frac{1}{S_{11}} \tag{A4}$$

For the pure nickel data, this gives $E_{\langle 100 \rangle} = 125 \text{ GPa}$. This is much less than the value for *polycrystalline* nickel, which, in the absence of texture, is expected to be isotropic, since the many grains average out the elastic properties over the different orientations. In fact, polycrystalline nickel has an elastic modulus of about 207 GPa, about 66% higher than that along the $\langle 100 \rangle$ direction. For nickel, it turns out that the $\langle 100 \rangle$ direction is the least stiff, and this is true for most other cubic metals.

How does one go about evaluating the elastic modulus at different crystallographic orientations, for example, along a direction with direction cosines l_1, l_2, l_3 defined relative to the 1, 2, 3 axes? To do this it is necessary to recognise that S is a fourth-rank tensor. Making use of the reduced notation, the new value of S_{11} , denoted S'_{11} , is given by

$$S'_{11} = \begin{bmatrix} +l_1^4 S_{11} & +2l_1^2 l_2^2 S_{12} & +2l_1^2 l_3^2 S_{13} & +2l_1^2 l_2 l_3 S_{14} & +2l_1^3 l_3 S_{15} & +2l_1^3 l_2 S_{16} \\ & +l_2^4 S_{22} & +2l_2^2 l_3^2 S_{23} & +2l_2^3 l_3 S_{24} & +2l_1 l_2^3 S_{25} & +2l_1 l_2^3 S_{26} \\ & & +l_3^4 S_{33} & +2l_2 l_3^3 S_{34} & +2l_1 l_3^3 S_{35} & +2l_1 l_2 l_3^2 S_{36} \\ & & & +l_2^2 l_3^2 S_{44} & +2l_1 l_2 l_3^2 S_{45} & +2l_1 l_2^2 l_3 S_{46} \\ & & & & +l_1^2 l_3^2 S_{55} & +2l_1^2 l_2 l_3 S_{56} \\ & & & & & +l_1^2 l_2^2 S_{66} \end{bmatrix} \tag{A5a}$$

for the general case of a monoclinic crystal. Note that the various 2's are present due to the symmetry of S . For a cubic crystal such as Ni, $S_{11} = S_{22} = S_{33}$, $S_{44} = S_{55} = S_{66}$ and $S_{12} = S_{13} = S_{23}$ with the other terms zero. Hence,

$$S'_{11} = S_{11} (l_1^4 + l_2^4 + l_3^4) + 2S_{12} (l_1^2 l_2^2 + l_1^2 l_3^2 + l_2^2 l_3^2) + S_{44} (l_1^2 l_2^2 + l_1^2 l_3^2 + l_2^2 l_3^2) \tag{A5b}$$

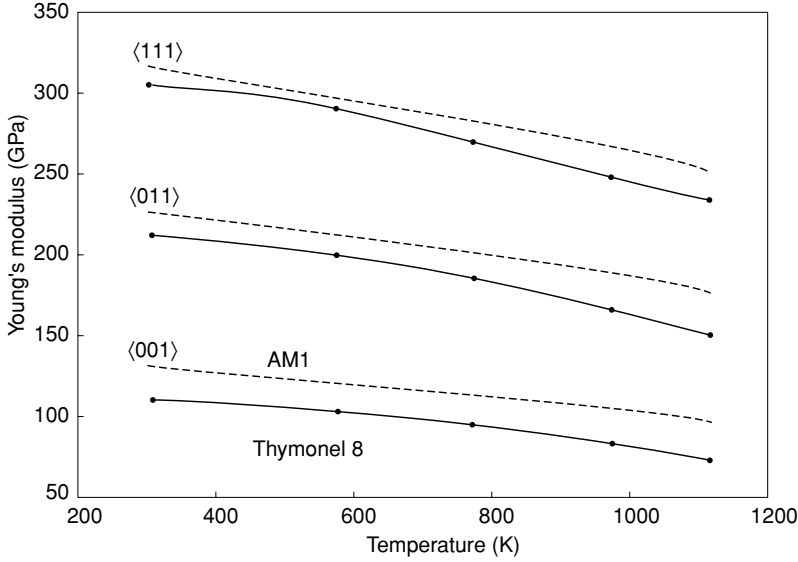


Fig. 2.75. Variation of the elastic modulus of single-crystal Thymonel and AM1 with crystallographic orientation and temperature [92].

Making use of the identities $l_1^2 + l_2^2 + l_3^2 = 1$ and $l_1^4 + l_2^4 + l_3^4 = 1 - 2(l_1^2 l_2^2 + l_1^2 l_3^2 + l_2^2 l_3^2)$, the transformation to the new axis is accomplished using

$$\frac{1}{E} = S'_{11} = S_{11} - 2 \left(S_{11} - S_{12} - \frac{1}{2} S_{44} \right) [l_1^2 l_2^2 + l_2^2 l_3^2 + l_3^2 l_1^2] \quad (\text{A6})$$

Thus, the degree of anisotropy depends upon the term $[l_1^2 l_2^2 + l_2^2 l_3^2 + l_3^2 l_1^2]$, which has a minimum value of zero along the cube axes, a maximum value of one-third along $\langle 111 \rangle$ and a value of one-quarter along $\langle 110 \rangle$. Using the values of S_{ij} for pure nickel, one can show that the $\langle 100 \rangle$ direction is the least stiff, $\langle 111 \rangle$ the stiffest (about 2.4 times stiffer than $\langle 100 \rangle$), with $\langle 110 \rangle$ between the two limits. The values are $E_{\langle 100 \rangle} = 125$ GPa, $E_{\langle 110 \rangle} = 220$ GPa and $E_{\langle 111 \rangle} = 294$ GPa. Single-crystal superalloys display a similar degree of anisotropy; see Figure 2.75 [92].

What can be said about the resistance to shearing, for example, in the 23 plane, which has as its normal the solidification direction $\langle 100 \rangle$? From the elasticity equations, one can see that this is largely controlled by the value of S_{44} ; however, the resistance to shearing does vary with angle in the 23 plane. One can transform S_{44} to S'_{44} using similar methods used to determine S'_{11} above; one can show that the rotation through an arbitrary angle θ is accomplished via

$$S'_{44} = S_{44} + (2S_{11} - 2S_{12} - S_{44}) \times 4 \sin^2 \theta \cos^2 \theta = S_{44} + (2S_{11} - 2S_{12} - S_{44}) \times \sin^2 2\theta \quad (\text{A7})$$

Thus, when $\theta = 0$, G is the reciprocal of S_{44} and about 119 GPa. When $\theta = \pi/4$, G is the reciprocal of S'_{44} and is about 26 GPa. For isotropic, polycrystalline nickel, the value of G has been determined to be about 82 GPa, not far from the numerical average of 119 GPa

and 26 GPa. More strictly, the value averaged over all loading configurations in the 23 plane is given by

$$\frac{4}{\pi} \int_0^{\pi/4} \frac{d\theta}{S_{44} + (2S_{11} - 2S_{12} - S_{44}) \times \sin^2 2\theta} = [2S_{44}(S_{11} - S_{12})]^{-1/2} \quad (\text{A8})$$

For the nickel data, this gives about 73 GPa. This is the appropriate value for columnar-grained nickel grown by directional solidification, with the long axis of each grain equivalent to the $\langle 100 \rangle$ growth direction.

For polycrystalline alloys, such as wrought superalloys or equiaxed castings, the elastic properties are expected to be isotropic and thus invariant with crystallographic orientation. Examination of Equation (A5) reveals that the condition for isotropy is

$$S_{11} - S_{12} - \frac{1}{2}S_{44} = 0 \quad (\text{A9})$$

In this case, the Young's modulus, E , Poisson's ratio, ν , and shear modulus, G , are given by the identities

$$S_{11} = \frac{1}{E} \quad S_{12} = -\nu/E \quad \text{and} \quad 2(S_{11} - S_{12}) = 1/G \quad (\text{A10})$$

Since $S_{44} = 2(S_{11} - S_{12})$, only two of E , ν and G are independent; the third can be calculated from a knowledge of the other two, for example, via identities such as $G = E/(2(1 + \nu))$. Measurements on polycrystalline nickel at room temperature have yielded $E = 207$ GPa, $\nu = 0.26$, and thus $G = 82$ GPa. For polycrystalline nickel one has therefore $S_{11} = 4.83 \times 10^{-12} \text{ m}^2/\text{N}$, $S_{12} = -1.26 \times 10^{-12} \text{ m}^2/\text{N}$ and $S_{44} = 12.17 \times 10^{-12} \text{ m}^2/\text{N}$.

Given the above considerations, it is interesting to determine whether the *elastic anisotropy* of nickel is greater or less than that displayed by other cubic metals. It is helpful to define an 'anisotropy factor', A , according to

$$A = \frac{2C_{44}}{C_{11} - C_{12}} \quad (\text{A11})$$

which will be unity in the case of isotropy. The value for Ni is 2.44, which compares with Cu at 3.19, Ag at 2.97, Au at 2.90, Ge at 1.65, Si at 1.57 and Al at 1.22, for other cubic metals. Thus, whilst Ni displays a fair degree of anisotropy, it is not the most anisotropic of FCC metals.

To summarise, Ni in single-crystal form is elastically anisotropic. When designing single-crystal blading from the nickel-based superalloys, it is important to recognise this fact. Compare, for example, the elastic properties of polycrystalline and directionally solidified nickel, which displays a strong $\langle 100 \rangle$ growth texture. Polycrystalline nickel displays values of E and G of 207 GPa and 82 GPa, respectively. Directionally solidified nickel displays a stiffness in the growth direction, denoted $E_{\langle 100 \rangle}$ of 125 GPa, i.e. only 60% of the isotropic value. The shear modulus for shearing in the plane whose normal is the growth direction is 73 GPa, about 89% of the isotropic case.

Questions

- 2.1 Nickel has an atomic weight of 58.71, a number which arises from the relative proportions of isotopes of weights 58, 60, 61, 62 and 64. Why is there little contribution from the isotopes of weight 59 and 63?
- 2.2 Nickel exhibits the FCC crystal structure, with a well-defined melting temperature of 1455 °C. But suppose that it was possible to supercool nickel liquid rapidly below the melting temperature, suppressing the formation of γ . At what approximate temperature might one expect the metastable HCP solid to form? Answer this question by investigating a binary phase diagram Ni–X, where X is an element from the d block which displays the HCP crystal structure. Given that the latent heat of melting of nickel is about 10 J/(mol K), estimate the difference in the Gibbs energy of the FCC and HCP forms at the temperature of solidification of the metastable HCP form.

Kaufman has carried out pioneering work on the lattice stabilities of the pure elements. His data for nickel (relative to the ferromagnetic FCC form) are: (i) liquid: $17\,614 - 10.299\,T$; (ii) HCP: $1046 - 1.255\,T$; (iii) BCC: $5564 - 1.06\,T$, where T is in Kelvin and the expressions give data in J/mol. Plot out these expressions as a function of temperature. Are Kaufman's data in reasonable agreement with your findings?

- 2.3 Consider a hypothetical binary phase diagram which displays complete solid solubility at low temperatures and a (solid + liquid) two-phase region below the liquid phase field. If both phases are ideal, demonstrate that just four parameters are sufficient for the calculation of the phase diagram. Using a spreadsheet, investigate how the shape and width of the two-phase region depends upon different combinations of the two parameters $\Delta S_1^{L \rightarrow \phi}$ and $\Delta S_2^{L \rightarrow \phi}$. Give a physical explanation for your findings.

In the Ni–Cr binary system, the liquid phase shows considerable deviation from ideality. For example, the Gibbs free energy of mixing at $x_{\text{Cr}} = 0.5$ is -13.1 kJ/mol at 1873 K. Using this information, modify the spreadsheet developed to describe the thermodynamic equilibria displayed by the Ni–Cr phase diagram. Demonstrate that the phase diagram is reproduced reasonably accurately.

- 2.4 Saunders has developed a thermodynamic model for the γ/γ' equilibria in the binary Ni–Al system. The molar Gibbs energy, $G_m^\phi\{T\}$, of each phase, ϕ , as a function of temperature (T) is represented by

$$G_m^\phi\{T\} - \sum_i x_i^\phi {}^0H_i\{298.15\text{ K}\} = G_m^{\text{ref},\phi} + G_m^{\text{ent},\phi} + G_m^{\text{E},\phi}$$

where x_i^ϕ is the mole fraction of component i in phase ϕ and ${}^0H_i\{298.15\text{ K}\}$ is the molar enthalpy of i in its standard state at 298.15 K. The terms $G_m^{\text{ref},\phi}$, $G_m^{\text{ent},\phi}$ and $G_m^{\text{E},\phi}$ are the reference, configurational entropy and excess contributions to the molar Gibbs energy. The disordered γ phase is modelled with a single sub-lattice, with

$$G_m^\gamma = x_{\text{Al}}^\gamma [G_{\text{Al}}^\gamma\{T\} - {}^0H_{\text{Al}}\{298.15\text{ K}\}] + x_{\text{Ni}}^\gamma [G_{\text{Ni}}^\gamma\{T\} - {}^0H_{\text{Ni}}\{298.15\text{ K}\}]$$

$$G_m^{\text{ent},\gamma} = RT [x_{\text{Al}}^\gamma \ln \{x_{\text{Al}}^\gamma\} + x_{\text{Ni}}^\gamma \ln \{x_{\text{Ni}}^\gamma\}]$$

and

$$G_m^{E,\gamma} = x_{Al}^\gamma x_{Ni}^\gamma L_{Al,Ni}^\gamma$$

where the term $L_{Al,Ni}^\gamma$ is an interaction parameter. The Ni_3Al phase can be modelled as a point compound, where

$$G_m^{\gamma'} = G_{Ni:Al}^{\gamma'}$$

such that there are no entropy and excess contributions.

Given the thermodynamic data given below, plot out the variation of G_m^γ and $G_m^{\gamma'}$ with temperature. Using graphical means or otherwise, determine the position of the $\gamma/(\gamma + \gamma')$ phase diagram as a function of temperature. Furthermore, estimate the ordering energy as a function of temperature.

The following thermodynamic data are in J/mol:

$$G_{Al}^\gamma = -11\,278.378 + 188.684\,153\,T - 31.748\,192\,T \ln T - 1.231 \times 10^{28} T^{-9};$$

$$G_{Ni}^\gamma = -5179.159 + 117.854\,T - 22.096\,T \ln T - 0.004\,840\,7\,T^2$$

$$L_{Al,Ni}^\gamma = -168\,750 + 16\,T + 30\,600(x_{Al}^\gamma - x_{Ni}^\gamma) + 41\,700(x_{Al}^\gamma - x_{Ni}^\gamma)^2$$

$$G_{Ni:Al}^{\gamma'} = 0.75\,G_{Ni}^\gamma - 40\,000 + 3\,T + 0.25\,G_{Al}^\gamma$$

- 2.5 Using published ternary phase diagrams for the Ni–Al–Cr system, investigate the effect on the solid-state equilibria of increasing the Cr content from zero to 25 at%, in a series of alloys containing (i) a constant Al content of 15 at% and (b) a constant Ni content of 75 at%. Use Thermo-Calc to help validate your conclusions.
- 2.6 Examination of Ni–Al–X ternary phase diagrams, where X is a ternary addition such as Ti, Nb, Pt, Re, etc., indicates that the γ' phase field is often parallel to either (i) the Ni–X axis or alternatively (ii) the Al–X axis. What causes this to happen? Very rarely, this is not the case: for Cr, for example. Why should such elements not conform this to this general rule?
- 2.7 In Figure 2.10 (taken from ref. [8]) the enthalpies of formation of the intermetallic compounds in the Ni–Al system are compared with the enthalpy of mixing of the FCC phase. Since the data correspond to the low temperature of 25 °C, it is claimed that the entropy contribution is small, so that ΔH_f behaves like ΔG_f .

Using the Thermo-Calc software, investigate whether this is a good assumption. Estimate a value of (i) the ordering energy of β –NiAl and (ii) the enthalpy of formation at 25 °C, measured with respect to the disordered BCC phase.

It is found that, of all the intermetallic phases in the Ni–Al system, the ordering energy is largest for β –NiAl. Remembering that this displays the CsCl crystal structure, give a physical explanation for this.

- 2.8 The lattice parameters of the γ and γ' phases in the superalloys depend upon the alloy chemistry. However, the lattice parameter of γ is, in general, slightly more sensitive to solute additions than γ' ; see Figure 2.18. Give a rationale for this observation.
- 2.9 The atom probe can be used to build up a ladder diagram in which the total number of atoms detected is plotted against the total number of atoms detected. For a specimen

- of pure Ni_3Al , explain the features of the curve which are observed, and estimate the slopes of the different regimes.
- 2.10 The lattice parameter of the Ni_3Al phase at room temperature is 0.357 00 nm. Given that it displays the L1_2 structure, estimate the atomic radii of Ni and Al in this compound. Is there any evidence that the bonding is directional in nature? How does the value for Ni compare with that of pure Ni, which has a lattice parameter of 0.351 67 nm?
 - 2.11 Investigate how the radius of the transition group metals varies as one crosses any one period of the periodic table. Repeat this exercise so that you have information for the first three series. What patterns emerge? And do the γ' -forming elements display significantly different values from the elements which form solid solutions with γ -nickel?
 - 2.12 In the intermetallic compounds γ' - Ni_3Al and β - NiAl , various sorts of point defects are possible. Distinguish between the four most common of these.
 - 2.13 Whilst both Frenkel and Schottky defects can arise in compounds such as β - NiAl , only one of these can occur in a pure elemental crystal such as Ni. Identify which of these defects may form in pure Ni, and explain why this is not possible for the other defect.
 - 2.14 Bradley and Taylor [41] made measurements of the lattice spacing and density of the NiAl intermetallic compound at compositions at and around the point of exact stoichiometry (Figure 2.76). Use their data to prove conclusively that (i) on the Ni-rich side, the excess Ni atoms are accommodated as anti-site defects, and (ii) on the Al-rich side, the excess Al atoms are not accommodated as anti-site defects and that there must be a significant proportion of constitutional vacancies present. What is the fraction of constitutional vacancies present for an alloy on the phase boundary on the Al-rich side?
 - 2.15 Theoretical expressions for the self-diffusion of pure metal indicate that the rate is proportional to Z , the number of neighbours nearest to any given atom. For nickel, which possesses the cubic F structure, determine Z and the corresponding lattice vector. How would your answers differ if nickel exhibited (i) the cubic P and (ii) the cubic I lattices?
 - 2.16 Explain why the tracer diffusion coefficient of Al in the γ' phase (which has the L1_2 structure) is considerably smaller than that of Ni.
 - 2.17 Comparison of the rates of interdiffusion of the transition group metals (the solutes) with nickel (the solvent) indicates that (i) the interdiffusion rate increases with increasing misfit strain between solvent and solute and (ii) the activation energy for interdiffusion decreases with increasing misfit strain. Why might these observations be contrary to expectation? How might this apparent anomaly be rationalised?
 - 2.18 The interdiffusion coefficient of nickel with aluminium in the β - NiAl intermetallic compound is strongly dependent upon concentration. The minimum value is near the point of 50:50 stoichiometry, but not quite so. Why is this?
 - 2.19 Self-diffusion in nickel is mediated by the exchange of atoms and vacancies on the lattice – so-called ‘vacancy-assisted diffusion’. The micromechanism involves the

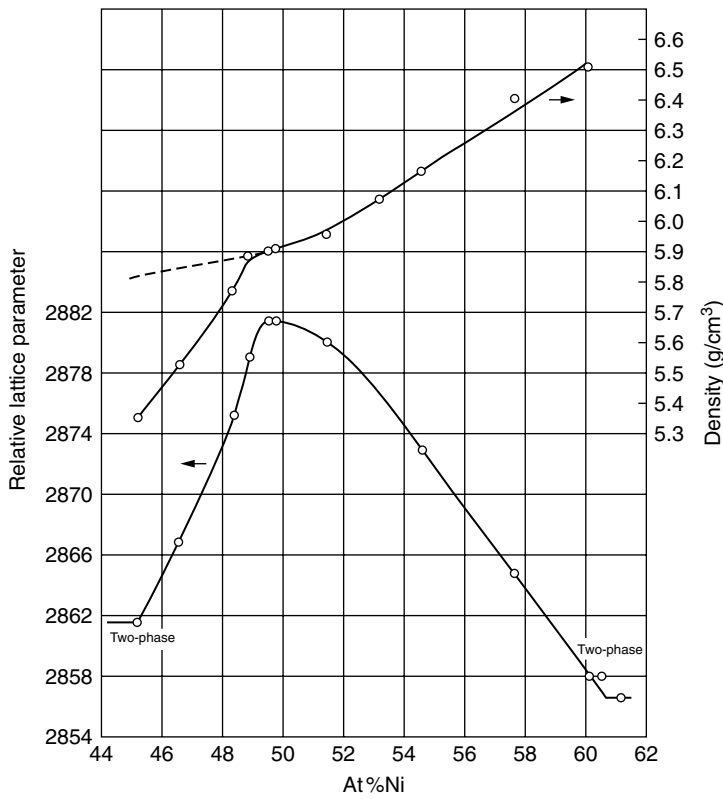


Fig. 2.76. Variation of the density and relative lattice parameter of NiAl around the point of stoichiometry [41].

movement of an atom by a distance $a/\sqrt{2}$ in the direction $\langle 110 \rangle$. Show that the size of the diffusion window is $(\sqrt{3} - 1)D_{\text{Ni}}$ at the saddle point, where D_{Ni} is the hard-sphere diameter of a Ni atom.

- 2.20 A rod of Ni is heated to its melting point, at which the concentration of vacancies is 3.3×10^{-5} per atomic site. The rod is then quenched so that all vacancies generated at the melting point are trapped on the lattice. Upon annealing at a higher temperature, the rod is observed to shrink. Why? If the total contraction is 5.1×10^{-6} , what change in lattice parameter should be observed during the annealing process?
- 2.21 Show that it is favourable for a perfect dislocation with Burgers vector $\frac{a}{2}[110]$ slipping on $(\bar{1}\bar{1}1)$ in a cubic close-packed (CCP) metal such as nickel to dissociate into two partial dislocations. Give the Burgers vectors of the partial dislocations. What factors determine their equilibrium separation?

Cobalt is often added to the nickel-based superalloys. Explain why the addition of cobalt is beneficial. Why might the effects of other elements be more potent?

- 2.22 Show that, on the basis of dislocation mechanics and isotropic elasticity, a perfect screw dislocation gliding in nickel will dissociate into two Shockley partial

dislocations separated by a distance $\frac{Ga^2}{48\pi\gamma} \left(\frac{2-3\nu}{1-\nu} \right)$, where G is the shear modulus, ν is Poisson's ratio and a is the lattice parameter.

The following information is provided. The shear stress field a distance r from the centre of a perfect screw dislocation of Burgers vector b is $Gb/(2\pi r)$. The associated elastic strain energy of this dislocation integrates to $Gb^2 \ln\{R/r_0\}/(4\pi)$, where r_0 is the core radius and R is the upper limit of integration.

- 2.23 The anti-phase boundary (APB) energy of the $L1_2$ structure is proportional to the ordering parameter, Ω , which is equal to $\Omega_{\text{NiAl}} - \frac{1}{2}(\Omega_{\text{NiNi}} + \Omega_{\text{AlAl}})$. By sketching sections of the (100), (011) and (111) planes, and noting that the lattices across the APB are related by a $1/2[01\bar{1}]$ vector, convince yourself (following Flinn) that (i) the energies of APBs on the (011) and (111) planes are in the ratio $\sqrt{3} : \sqrt{2}$, and (ii) the energy of an APB on the (100) plane is likely to be nearly zero. Consider only the nearest-neighbour interactions.
- 2.24 Stobbs has made measurements of the APB energies in the disc alloy RR1000, using weak-beam TEM microscopy to image dislocation pairings in the γ' phase. His figure for the {100} plane, 200 mJ/m^2 , is not very different from that in the {111} plane, at 300 mJ/m^2 . This is despite Flinn's analysis (see Question 2.23), which indicates that the {100} APB should be close to zero. How can this apparent anomaly be resolved?
- 2.25 The Nb-rich compound Ni_3Nb , known as γ'' , confers precipitation strengthening to iron-rich superalloys such as IN718. It possesses a body-centred tetragonal crystal structure, which is related to that of γ' by a somewhat different ordering scheme (see Figure 2.21) and a doubling of the lattice parameter in one of the cube directions.

Draw out a plan of a pseudo-close-packed plane of the γ'' phase, and identify the magnitudes of possible Burgers vectors. Hence convince yourself that in only one of three directions will order be restored as a result of the movement of two $a/2\langle 110 \rangle$ matrix dislocations cutting through γ'' , and that, in the other two directions, four such dislocations are required. Thus explain why γ'' -strengthened alloys possess extremely high strength levels at elevated temperatures.

- 2.26 Oblak, Paulonis and Duvall [93] studied the flow characteristics of the iron–nickel superalloy IN718, in single-crystal form, with the tensile axis along $\langle 001 \rangle$. When tested at 300°C , the yield stress was 300 MPa ; however, if the specimens were first aged at 790°C for 5 h, the yield stress was altered by $+25\%$ and -6% for tensile and compressive loads, respectively. Give an explanation for these findings.
- 2.27 Each of the alloying elements present in a superalloy is expected to make a contribution to the level of solid-solution strengthening, such that the yield stress is increased by an increment $\Delta\sigma$. To sum the contributions from the concentrations, c_i , of each element i , it has been suggested that one uses

$$\Delta\sigma = \left(\sum_i k_i^2 c_i \right)^{1/2}$$

Tensile testing at room temperature indicates that suitable values for the k_i 's are Al = 225, Ti = 775, Zr = 2359, Hf = 1401, Cr = 337, Mo = 1015, W = 977, Mn = 448, Fe = 153 and Co = 39, where the units are in $\text{MPa} \times \text{at.fraction}^{-1/2}$.

Use this relationship and the data provided to quantify, for the single-crystal superalloy SRR99, the relative contributions of the different elements to the solid-strengthening levels to be expected in this alloy.

- 2.28 In order to evaluate the temperature dependence of the shear yield stress, τ_y , of a solid-solution-strengthened alloy, one can appeal to the trough model, which predicts

$$T\left(\frac{\tau_y}{G}\right)^{1/2} = K_1 - K_2\tau_y$$

where T is temperature and G is the shear modulus. The terms K_1 and K_2 are constants, which are in the range 5.7 to 27 K (K_1) and 0.22 to 0.26 K/MPa (K_2).

Using the expression, the data provided and a suitable estimate for G , investigate how the shear yield stress, τ_y , is expected to vary as a function of temperature. What is the effect of incorporating a temperature dependence into G ?

- 2.29 The expression derived in Section 2.3 for the strengthening arising from strongly coupled dislocation pairs is dependent upon the function $\ell\{x\}$. For spherical particles of radius r , $\ell\{x\}$ is given by $2(2rx - x^2)^{1/2}$; the results in a $r^{-1/2}$ dependence for the strengthening, when r is large.

Suppose instead one considers cuboidal particles. What then is the appropriate shape of the cross-section, if the slip plane is $\{111\}$? Derive an expression for $\ell\{x\}$ for this case. Is the $r^{-1/2}$ dependence preserved? If not, is the dependence on r stronger or weaker than for spherical particles?

- 2.30 In nominally pure nickel in the annealed state, the movement of dislocations during plastic deformation is not limited by the Peierls–Nabarro stress, which is of very small magnitude (in common with all FCC materials). Evidence from this comes from TEM studies, which indicate that the dislocations are wavy with little evidence of pinning being displayed. This being the case, what phenomenon controls dislocation movement? How would the situation be different if nickel were to be heavily worked?
- 2.31 The shear strain rate, $\dot{\gamma}$, arising due to low-temperature plasticity has been modelled in a phenomenological sense by Kocks, Argon and Ashby according to

$$\dot{\gamma} = \dot{\gamma}_0 \left(\frac{\sigma_s}{\mu}\right)^2 \exp \left\{ -\frac{\Delta F}{kT} [1 - (\sigma_s/\tau_p)^p]^q \right\}$$

where σ_s is the deviatoric stress, μ is the shear modulus, τ_0 is the shear stress at 0 K and ΔF is an activation energy required either for (i) overcoming discrete obstacles or else (ii) formation of an isolated pair of kinks. The other terms, $\dot{\gamma}_0$, p and q , are constants; theory suggests that $0 \leq p \leq 1$ and $1 \leq q \leq 2$, with $\dot{\gamma}_0$ about 10^6 /s.

Make a fit to the flow stress–temperature data for nickel (see Figure 2.52), including data down to cryogenic temperatures. Do your fitted parameters reveal much about the factors controlling slip? What is revealed about the strain-rate sensitivity? Convince yourself that the strain-rate sensitivity is low compared with, for example, that seen during power-law creep.

- 2.32 The elastic modulus of nickel is highly anisotropic; the Young's moduli along $\langle 111 \rangle$, $\langle 110 \rangle$ and $\langle 100 \rangle$ at ambient temperature are in the ratio 2.80:1.71:1. By determining the interplanar spacings in the cubic close-packed (CCP) crystal structure, rationalise

the observation that the $\langle 111 \rangle$ and $\langle 100 \rangle$ directions are the most and least stiff, respectively.

- 2.33 The variation of the Young's modulus around the $\langle 100 \rangle$ pole is of profound importance in the design of turbine blading, since resonance can cause high-cycle fatigue failure. Using the expressions given in the Appendix to this chapter, investigate how the Young's modulus varies within 20° of $\langle 100 \rangle$. For CMSX-4 blading at 700°C , take $E_{100} = 104\text{ GPa}$, $E_{110} = 190\text{ GPa}$ and $E_{111} = 262\text{ GPa}$.
- 2.34 In the context of the elasticity of monocrystalline nickel, prove the following identities:

$$S_{11} = \frac{C_{11} + C_{12}}{(C_{11} - C_{12})(C_{11} + 2C_{12})} \quad S_{12} = \frac{C_{12}}{(C_{11} - C_{12})(C_{11} + 2C_{12})} \quad S_{44} = 1/C_{44}$$

where the S_{ij} and C_{ij} are components of the compliance and stiffness matrices, respectively.

- 2.35 In Section 2.4, the σ^5 stress dependence of steady-state creep in pure nickel was justified on the basis that time scales with the third power of the characteristic length of the dislocation sub-structure λ . But suppose instead that a fifth power law (corresponding to coarsening of the sub-grains by pipe diffusion) applies. Demonstrate that the exponent in the steady-state creep law is then equal to 7.
- 2.36 In the climb-plus-glide creep model of Weertman, the total time, t_{total} , that a dislocation spends deforming is partitioned into glide and climb components, such that

$$t_{\text{total}} = t_{\text{glide}} + t_{\text{climb}}$$

By (i) dividing through by the product ρbL , where ρ is the dislocation density, b is the Burgers vector and L is the average distance that a dislocation moves parallel to the slip plane, and (ii) defining the total, glide and climb components of the strain rate in an appropriate way, demonstrate that

$$\frac{1}{\dot{\epsilon}_{\text{total}}} = \frac{1}{\dot{\epsilon}_{\text{glide}}} + \frac{1}{\dot{\epsilon}_{\text{climb}}}$$

so that one has a sequential process for the net strain rate.

- 2.37 Creep-rupture data for polycrystalline, nominally pure nickel are plotted on Figure 2.66. Use these data to plot a line for Ni on the Larson–Miller plot of Figure 1.16. Hence show that many intermetallics are superior to Ni in creep, but that they cannot compete at present with the single-crystal superalloys.

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3 Single-crystal superalloys for blade applications

The gas turbine consists of many different pieces of turbomachinery, but the rows of turbine blading are of the greatest importance since many engine characteristics, for example the fuel economy and thrust, depend very strongly on the operating conditions which can be withstood by them. Thus very arduous temperatures and stresses are experienced by the materials employed, which are pushed near to the limits of their capability. This is particularly the case for the high-pressure blades, which are located nearest to the hot gases emerging from the combustion chamber. Their function is to extract work from the gas stream and to convert it to mechanical energy in the form of a rotating shaft, which drives the high-pressure compressor.

A consideration of the operating conditions experienced by the high-pressure turbine blades in a large civil turbofan engine, such as the Rolls-Royce Trent 800 or General Electric GE90, confirms this point. The temperature of the gas stream is about 1750 K, which is above the melting temperature of the superalloys from which the blades are made. The high-pressure shaft develops a power of about 50 MW – hence, with about 100 blades, each extracts about 500 kW, which is sufficient to satisfy the electricity requirement of about 500 homes. Each row of blades is expected to last at least 3 years, assuming they operate at 9 h/day. This is equivalent to about 5 million miles of flight, or ~ 500 circumferences of the world. The blades rotate at an angular speed, ω , of more than 10 000 revolutions per minute, so that the tip velocity is greater than 1200 km/h. The stresses experienced by the blading, which are largely centrifugal in nature due to the significant rotational speeds, are considerable; if the blade cross-section is assumed to be uniform, the stress, σ , at the blade root is approximately given by

$$\sigma = \int_{r_{\text{root}}}^{r_{\text{tip}}} \rho \omega^2 r \, dr = \frac{\rho \omega^2}{2} (r_{\text{tip}}^2 - r_{\text{root}}^2)$$

where ρ is the blade density. The radial positions of the blade tip and root are given by r_{tip} and r_{root} , respectively. Taking $\rho = 9000 \text{ kg/m}^3$, the blade length as 10 cm and the mean blade radius to be 0.5 m, the stress at the blade root is given by

$$\sigma = 9000 \times (100 \times 2\pi)^2 \times (0.55^2 - 0.45^2)/2 = 180 \text{ MPa}$$

Since the cross-sectional area is, in general, no greater than a few square centimetres, this is equivalent to the weight of a heavy truck hanging on each blade. The mass of a shroud and coatings will add to the value calculated. These considerations confirm that rather special materials are required for this application.

It has become common to process turbine blading by investment casting of the superalloys, very often into single-crystal form. Not surprisingly, considerable attention has been paid to the optimisation of the compositions of the alloys employed. The mechanical properties of the alloys are of great importance. Furthermore, the nature of the application means that great care needs to be taken in the design of the shape of the turbine blading. These topics are considered in this chapter.

3.1 Processing of turbine blading by solidification processing

The blading for the very first gas turbine engines was produced by extrusion and forging operations [1]. However, in the 1970s it became apparent that there were severe limitations with this approach. First, the blades tended to be heavy, since it proved impossible to make them in hollow form, without subsequent machining operations. Second, they were prone to cracking and incipient melting, since high working temperatures were required, particularly for alloys with high yield stresses. Thus, nowadays, superalloy turbine blading is always produced by investment casting [2]; it has been estimated recently that this industry alone is worth about \$3 billion per annum. Hollow aerofoils with complicated shapes are produced routinely, incorporating internal cooling passages to allow cool air to flow inside the blade during operation, to restrict the metal temperatures to reasonable levels. A further advantage is that the grain boundaries can be eliminated, allowing the very best creep properties to be attained. This could not be achieved using thermomechanical processing.

3.1.1 *The practice of investment casting: directional solidification*

The investment casting or ‘lost-wax’ process [3,4] involves the following steps. First, a wax model of the casting is prepared by injecting molten wax into a metallic ‘master’ mould – if necessary (for hollow blades) by allowing wax to set around a ceramic core, which is a replica of the cooling passages required. These are arranged in clusters connected by wax replicas of runners and risers; this enables several blades to be produced in a single casting. Next, an investment shell is produced by dipping the model into ceramic slurries consisting of binding agents and mixtures of zircon (ZrSiO_4), alumina (Al_2O_3) and silica (SiO_2), followed by stuccoing with larger particles of these same materials. This operation is usually repeated three or four times until the shell thickness is adequate. Finally, the mould is baked to build up its strength. The first step involves a temperature just sufficient to melt out the wax – usually a steam autoclave is used. Further steps at higher temperatures are employed to fire the ceramic mould. After preheating and degassing, the mould is ready to receive the molten superalloy, which is poured under vacuum at a temperature of $\sim 1550^\circ\text{C}$. After solidification is complete, the investment shell is removed and the internal ceramic core leached out by chemical means, using a high-pressure autoclave. It is clear from this description that many steps are required. Fortunately, in most modern foundries considerable amounts of automation have been introduced. In Figure 3.1, the various steps in the investment casting process are summarised. A wax model, the resulting casting and the final machined high-pressure turbine blade are shown in Figure 3.2.

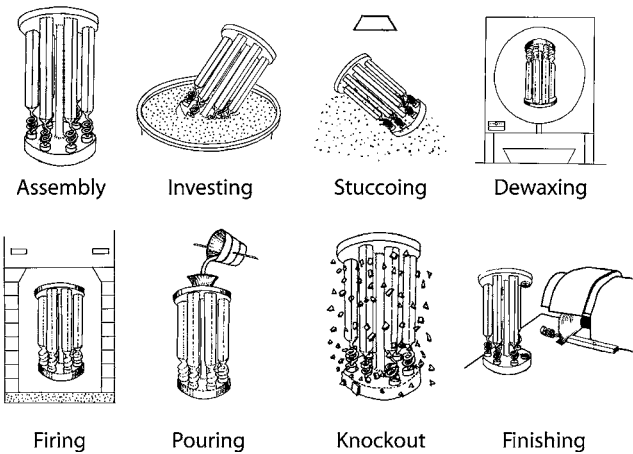


Fig. 3.1. Illustration of the various stages of the investment casting process. (Courtesy of Mr Frank Reed.)

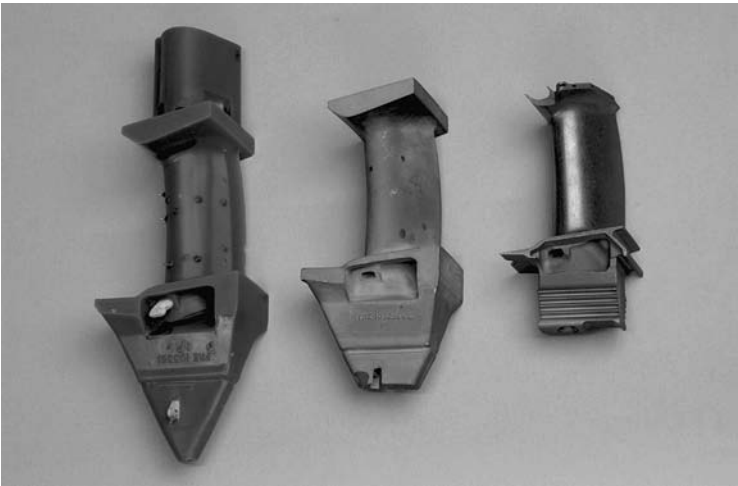


Fig. 3.2. The Trent 800 high-pressure turbine blade at different stages of its production by investment casting. Left: wax model containing ceramic core, ready to receive the investment shell. Centre: finished casting with pig-tail selector removed. Right: finished blade, after machining.

When investment casting was first applied to the production of turbine blading, equiaxed castings were produced by the ‘power-down’ method, which involved the switching off of the furnace after pouring of the molten metal [3]. However, it has been found that the creep properties are improved markedly if the process of *directional solidification* is used – these methods were pioneered by Versnyder and co-workers at the Pratt & Whitney company in the 1970s, leading to the introduction of directionally solidified blading cast from PWA1422 in the 1980s. After pouring, the casting is withdrawn at a controlled rate from the furnace – as in the conventional Bridgman crystal-growing method. A speed of a few inches per hour is typical – so that the solid/liquid interface progresses gradually along the casting, beginning at its base. This has the effect of producing large, columnar grains which are

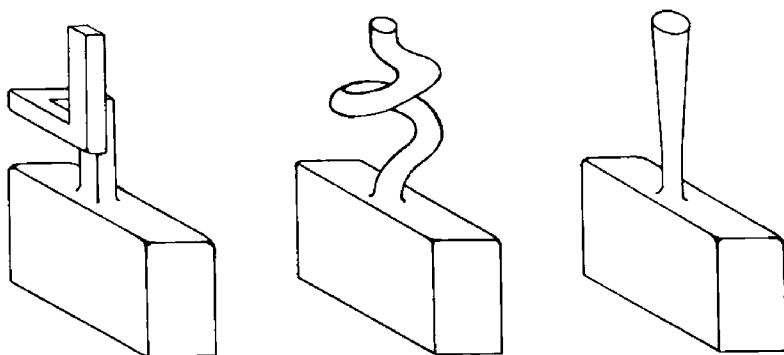


Fig. 3.3. Various designs of grain selector, for the production of the investment castings in single-crystal form.

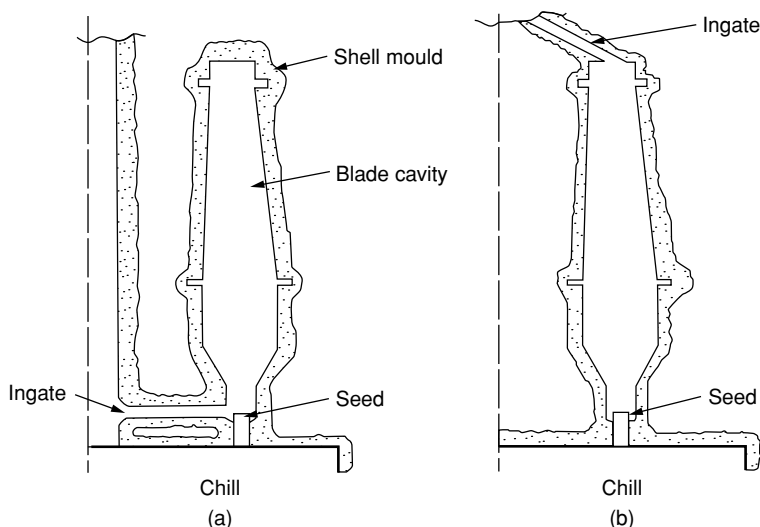


Fig. 3.4. Two configurations by which turbine blading can be grown in single-crystal form using the seeding technique. (a) Molten metal enters via a downpole, with an ingate at the base of the casting. (b) Downpole at the top in a so-called 'top-pouring' configuration.

elongated in the direction of withdrawal, so that transverse grain boundaries are absent [5]. In a variant of this process, the grain boundaries are removed entirely. Most typically, this is achieved [6] by adding a 'grain selector' to the very base of the wax mould, typically in the form of a pig-tail-shaped spiral; see Figure 3.3. Since this is not significantly larger in cross-section than the grain size, only a single grain enters the cavity of the casting, which is then in monocrystalline form. Alternatively, a seed can be introduced at the base of the casting (see Figure 3.4); provided the processing conditions are chosen such that this is not entirely remelted, growth occurs with an orientation consistent with that of the seed. In Figure 3.5, a high-pressure turbine blade casting is shown, with pig-tail grain selector still in place. Figure 3.6 shows a close-up of the base of the casting. Note, in particular, the columnar grain structure in the region below the grain selector.



Fig. 3.5. Rolls-Royce Trent 800 high-pressure turbine blade casting, after removal of the investment shell.



Fig. 3.6. Details of the starter block and pig-tail grain selector from the Trent 800 high-pressure turbine blade casting.

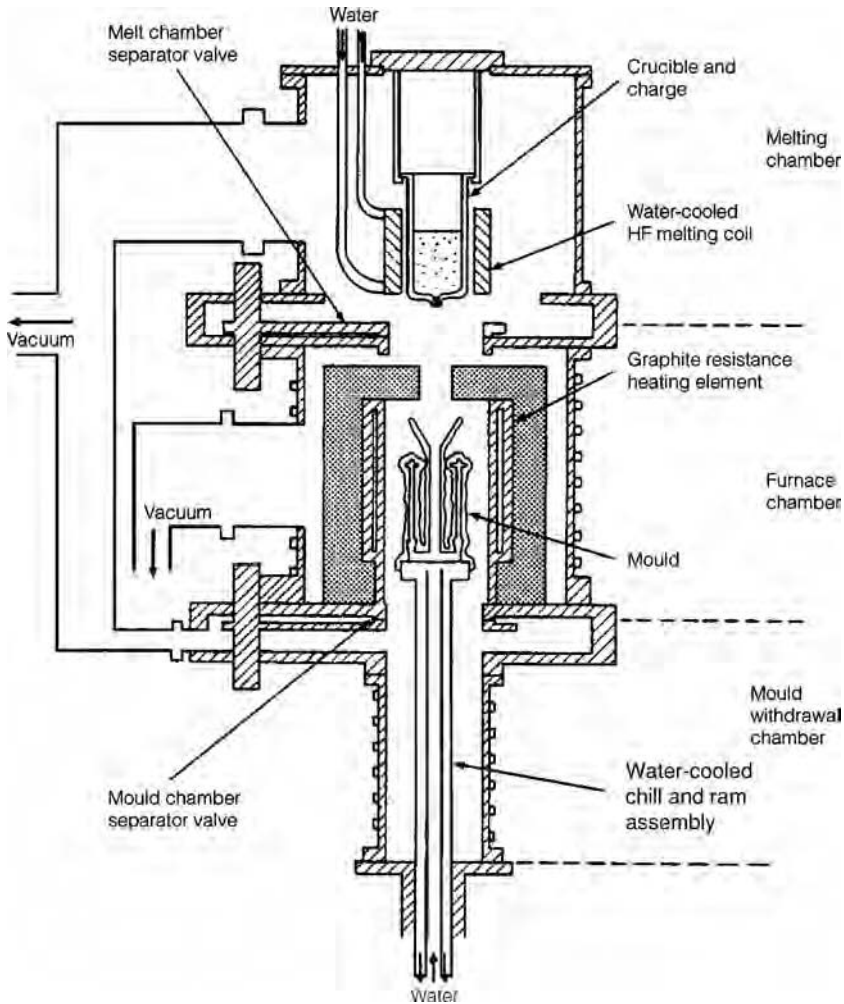


Fig. 3.7. Schematic illustration [6] of the investment casting furnace used by Rolls-Royce plc for the production of single-crystal turbine blading.

Figure 3.7 is a schematic illustration [6] of the casting furnace used for the production of turbine blading at Rolls-Royce plc, Derby, UK. The investment shell mould sits on a copper chill plate of about 14 cm diameter; this size is sufficient for five HP turbine blades for a civil jet engine to be produced simultaneously, or alternatively 20 for a smaller helicopter engine. A bottom-run mould system is chosen, which allows the use of integral ceramic filters to prevent ceramic particles from being swept into the blade cavities, which are then filled in a controlled, predictable fashion. Since the diameter of the chill plate is small, as little as 5 kg of charge needs to be melted. This is achieved, typically within 2 minutes, by using high-frequency (HF) induction melting in a disposable, refractory-fibre crucible. The furnace consists of an upper melting chamber, a central mould chamber and a lower withdrawal chamber, each of which are individually pumped and valved. Once a

suitable vacuum is achieved in all chambers, the mould is raised into the central chamber, which is maintained at a temperature above the liquidus by a graphite resistance heater. Immediately after pouring, the mould is withdrawn at the programmed rate. In practice, a high thermal gradient of typically 4000 K/m is achieved, which is sufficient to produce cast microstructures of dendritic form (see Figure 3.8); the primary dendrite arm spacing is then in the range 100–500 μm [7]. The cooling time between the critical temperatures of 1400 °C and 1000 °C is typically about 20 minutes.

In this way, it has proven possible to cast turbine blade aerofoils with complicated patterns of cooling channels. Over time, significant improvements have been made to the cooling configurations. For example, the early extruded and cast equiaxed HP blades introduced in the 1970s on the Rolls-Royce RB211-22B and early RB211-524s had single-pass cooling configurations; see Figure 3.9(a). The air had a relatively short path inside the blade before being exhausted, which limited its potential to absorb heat through contact with the metal; this was so despite the incorporation of extensive numbers of film-cooling holes, typically drilled using Nd–YAG lasers. The latest generation RB211 and Trent blades use a multi-pass, or ‘serpentine’, configuration (see Figure 3.9(b)), in which the flow passes through longer passages before being exhausted. This maximises the heat pick-up of the cooling air, enabling increases to the main stream gas temperatures and reductions in the required cooling flow. These improvements in cooling technologies have allowed in part the turbine entry temperature of a typical large turbofan engine to be increased by about 250 °C, at a conservative estimate (see Figure 1.5), with associated improvements in engine efficiency.

The fact that the creep properties are improved if columnar structures are produced implies that the grain boundaries are a source of weakness. This has been demonstrated very clearly with elegant experiments on the CMSX-4 superalloy, which is designed to be used in single-crystal form, i.e. with no grain boundaries present. Castings were fabricated with the presence of high-angle grain boundaries, using two seeds of known orientation – this allowed the creep life to be measured of a specimen containing a grain boundary defect of known misorientation θ . The results are given in Figure 3.10. The data indicate that for an applied stress of 300 MPa at 850 °C, a perfect single crystal has a stress rupture life in excess of 10 000 h; however, a grain boundary of misorientation $\theta = 7^\circ$ reduces the creep life to 100 h. When $\theta > 10^\circ$, the creep life is only a few hours. Clearly, the creep rupture life depends very sensitively on θ . This is one reason why so-called ‘grain-boundary-tolerant’ single-crystal superalloys have recently become available – these contain the grain-boundary strengtheners such as carbon and boron present in the early superalloys which were used for equiaxed castings. The sensitivity of the creep life to the presence of grain boundaries is supported by early work [8] on Mar-M200, one of the first alloys to be used in single-crystal form; see Figure 3.11. Under creep loading of about 200 MPa and 980 °C, the removal of the transverse grain boundaries increases the creep ductility from below 5% to in excess of 25%. In single-crystal form, the creep life is improved to greater than 100 h from about 70 h for the directionally solidified, columnar-grained structure. When grain boundaries are present in these systems, creep damage occurs in the form of voids and cavitation; see Figure 3.12. If grain-boundary-strengthening elements such as carbon are present, cracking of the carbide precipitates and their decohesion from grain boundaries is commonplace.

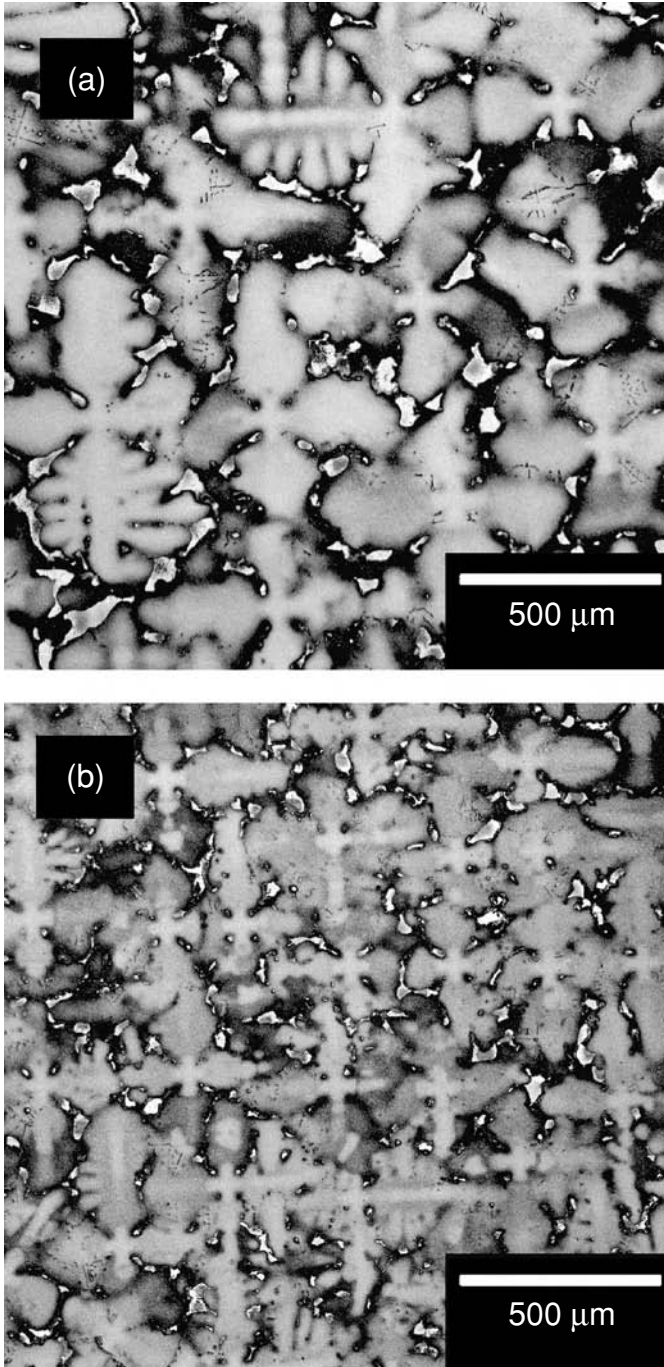


Fig. 3.8. Scanning electron micrographs [7] of the dendritic microstructure of a single-crystal superalloy, in the as-cast state. (a) 2.5 mm/min, conventional Bridgman furnace; (b) 2.5 mm/min with liquid tin as a metal coolant. (Courtesy of Andrew Elliott.)

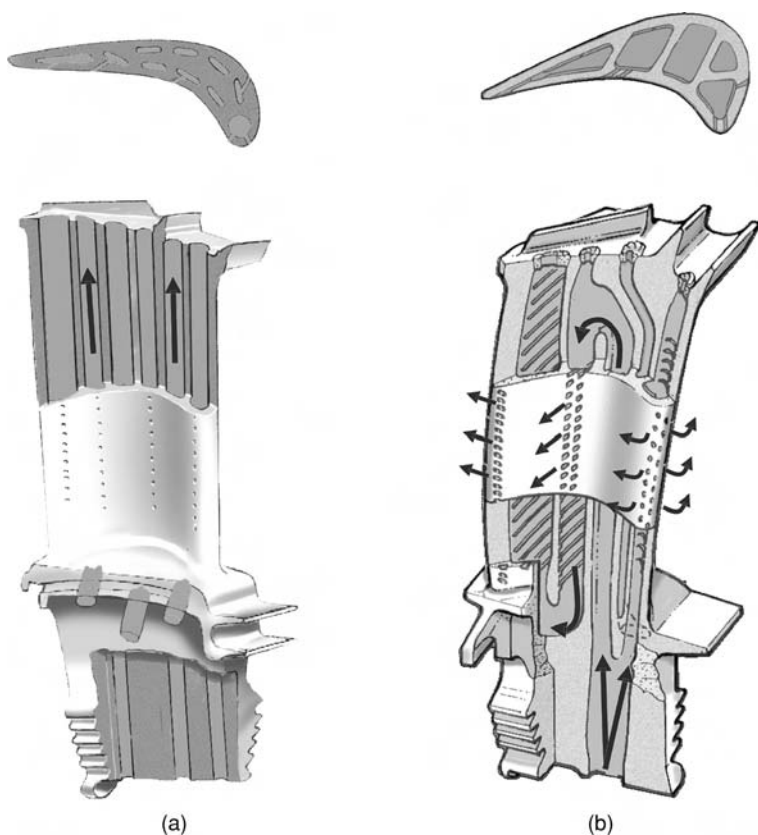


Fig. 3.9. Schematic illustration of the cooling configurations used for turbine blade aerofoils. (a) Single-pass cooling. (b) Multipass 'serpentine' cooling. In both cases, holes for film cooling are present. (Courtesy of Rolls-Royce.)

The removal of the grain boundaries using a grain selector, or else by seeding, has ramifications for the crystallographic orientation of the casting. The preferred growth direction for nickel and its alloys, in common with all known FCC alloys, is $\langle 001 \rangle$. Thus turbine blade aerofoils have a $\langle 001 \rangle$ direction aligned along, or close to, the axis of the casting. The texture arises as a consequence of the competitive growth of dendrites in the starter block of the casting [9]; see Figure 3.13. The solid nuclei which form adjacent to the chill block have an orientation which is random, but after only 1 mm of growth a strong $\langle 001 \rangle$ -orientated texture is present; see Figure 3.14. This effect occurs because the dendrites grow at a rate which is largely controlled by solute diffusion, since the solid phase grows from the liquid with a very different composition from it; thus the local dendrite tip undercooling, ΔT_{tip} , scales monotonically with the velocity, v , of the dendrite, measured along the temperature gradient. It follows that dendrites which are misaligned by an angle θ with respect to perfectly aligned ones must grow at a greater undercooling and hence at the rear of the growth front; see Figure 3.15. This provides an opportunity for the secondary and tertiary arms of the well aligned dendrites to suppress the primary arms of those which are less so. The net

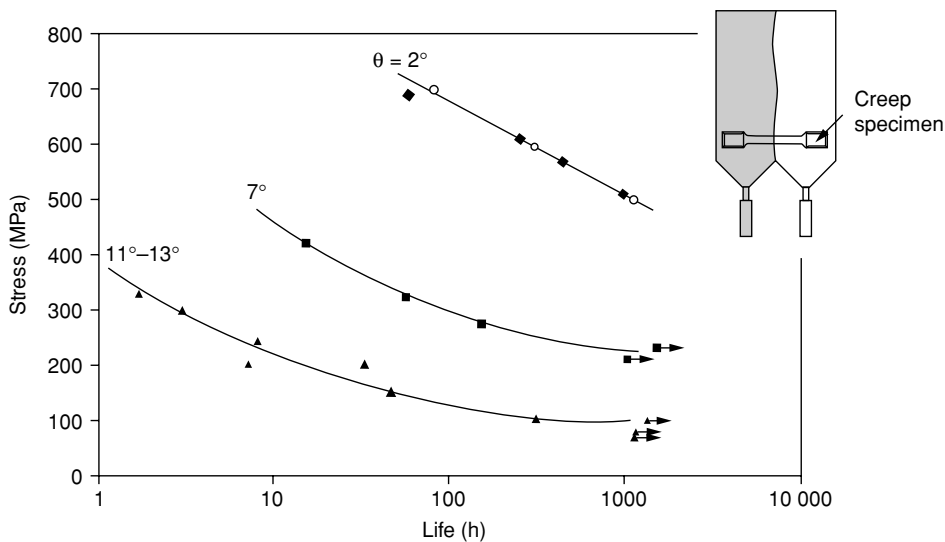


Fig. 3.10. Variation of the creep rupture properties of the superalloy CMSX-4 at 850 °C, measured on specimens with a single grain boundary of misorientation, θ , introduced. Note the very great reduction in performance caused by the presence of the grain boundaries. (Courtesy of Bob Broomfield.)

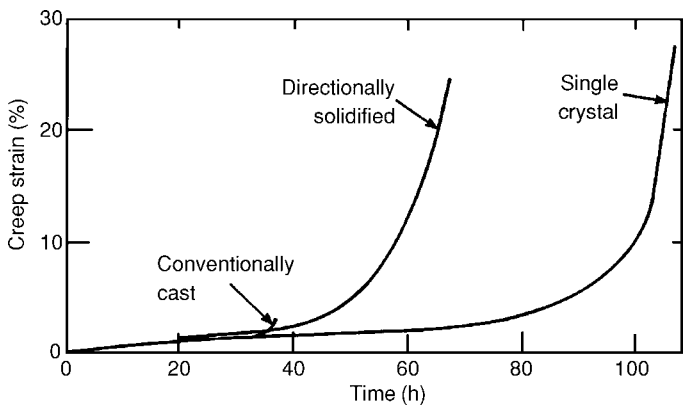


Fig. 3.11. Evolution of creep strain with time for the superalloy Mar-M200 under loading of 206 MPa and 982 °C, in the conventionally cast (equiaxed) state, the directionally solidified (columnar) state and the single-crystal form [8].

result is a reduction in the number of grain boundaries and the establishment of a strong $\langle 001 \rangle$ -orientated texture.

3.1.2 Analysis of heat transfer during directional solidification

In practice, directional solidification requires careful control of the rate of heat removal from the casting. Mathematical analysis of the heat transfer processes is therefore useful, since it provides insight into the effects occurring.

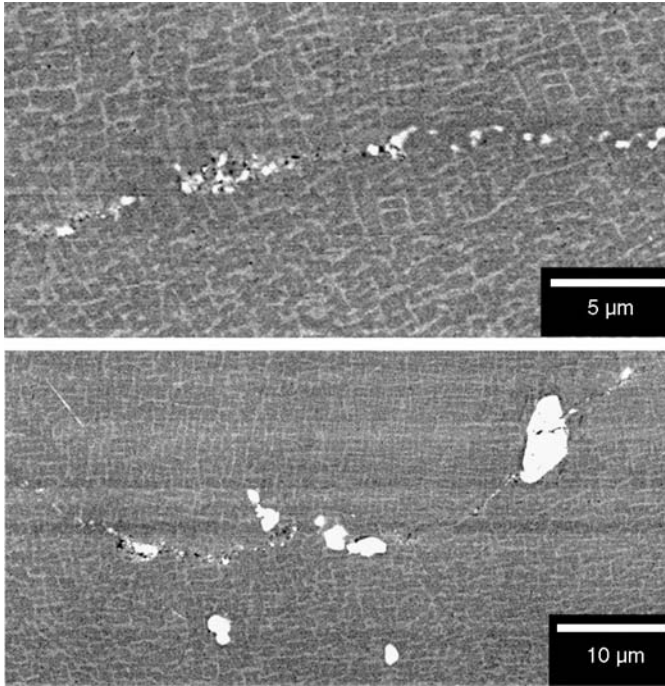


Fig. 3.12. Scanning electron micrographs of the grain boundaries in the directionally solidified superalloy RR2072, modified with 0.05 wt%C and 0.005 wt%B. (a) 18.8° misorientation, crept to 64 hours (80% of creep life); (b) 10.5° misorientation, crept to 100 hours (50% of creep life). (Courtesy of Quizi Chen.)

A Treatment of infinite rod – estimation of withdrawal velocity

One can make a first estimate of the *maximum* speed at which the casting should be removed from the furnace in the following way. As a first approximation, one treats the casting as an infinite, cylindrical rod; this moves at a uniform rate, v , from a furnace of temperature T_1 into a medium of ambient temperature T_0 ; see Figure 3.16. The heat flow is taken to be unidirectional, i.e. it occurs only in the axial direction (denoted x). The temperature field, $T\{x\}$, comes to a steady-state, which satisfies the heat flow equation in the moving frame,

$$\kappa \frac{\partial^2 T}{\partial x^2} + vc \frac{\partial T}{\partial x} + \dot{Q} = 0 \quad (3.1)$$

where $\dot{Q}$ is the local volumetric rate of heat production or extraction, c is the volumetric specific heat capacity and κ is the thermal conductivity. At the melting temperature, when $T = T_m$ (note $T_1 > T_m > T_0$), the *production* of latent heat occurs; this requires that the following boundary condition is satisfied at the solid/liquid interface:

$$\kappa_s \left. \frac{\partial T}{\partial x} \right|_s = \kappa_l \left. \frac{\partial T}{\partial x} \right|_l + Lv \quad (3.2)$$

where L is the volumetric latent heat of fusion and the subscripts s and l refer to the solid and liquid, respectively. Conversely, heat is *extracted* at a rate $q = h_{\text{eff}}(T - T_0)$ by Newtonian

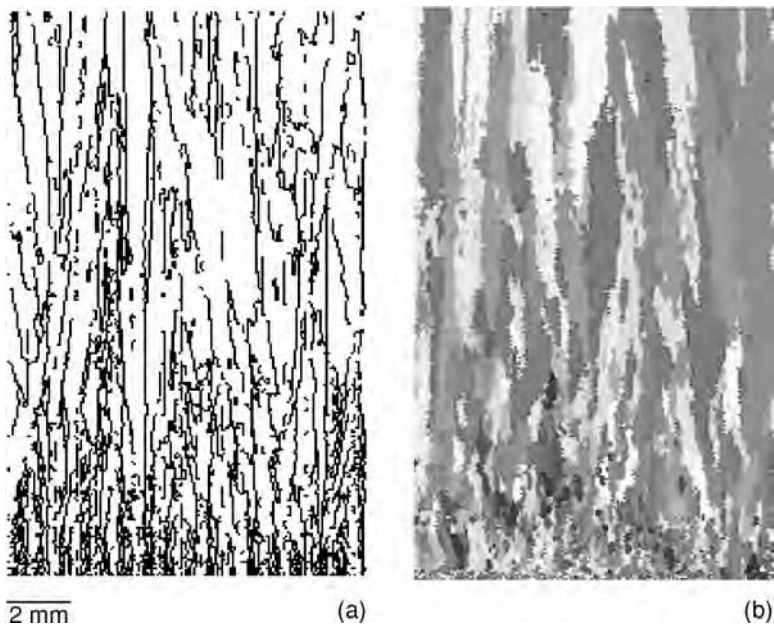


Fig. 3.13. Grain structure in the starter block of a single-crystal turbine blade casting, cast from CMSX-4; the chill block is at the bottom and the growth direction is upwards. (a) Results from theoretical calculations made using the phase-transformation theory. (b) Experimental analysis using electron back-scattered diffraction patterns of the starter block in Figure 3.6, confirming the evolution of the $\langle 001 \rangle$ texture. Data taken from ref. [9].

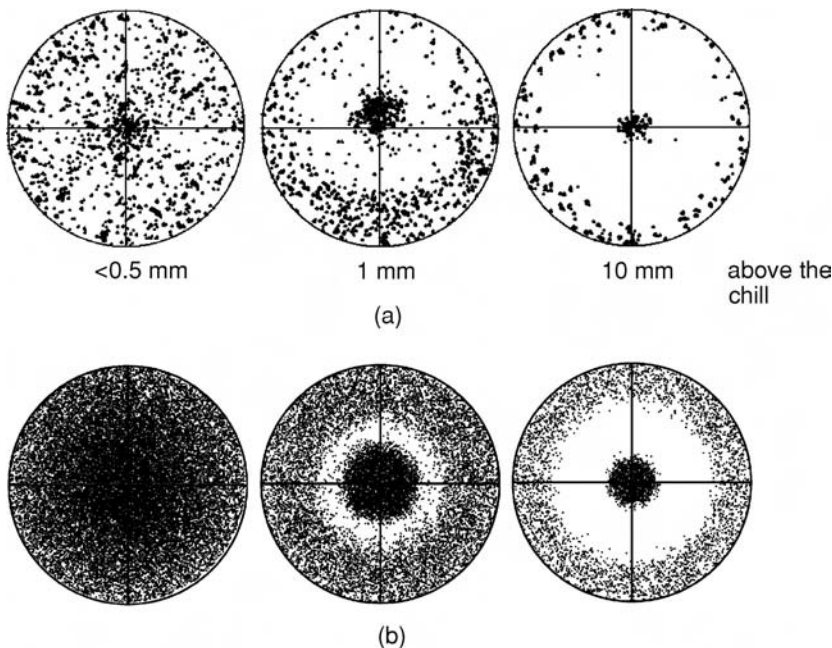


Fig. 3.14. (a) Measured and (b) predicted $\langle 001 \rangle$ pole figures taken from the starter block shown in Figure 3.6 at various heights above the chill block [9].

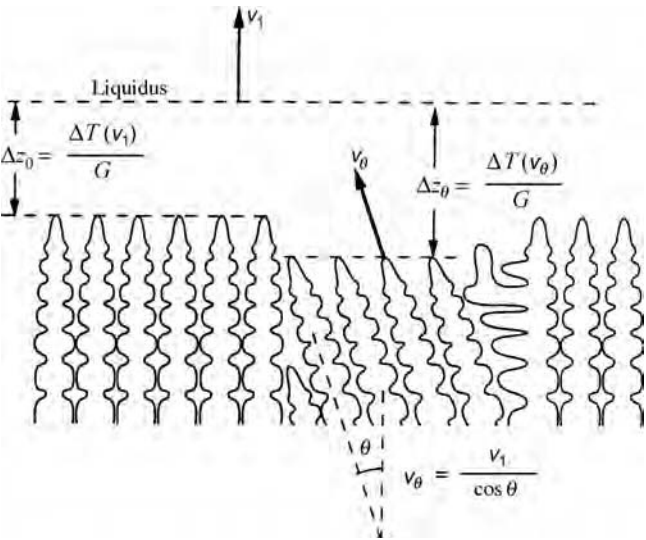


Fig. 3.15. Illustration of the competitive grain growth process, in which misaligned dendrites are suppressed by the secondary arms of well aligned ones.

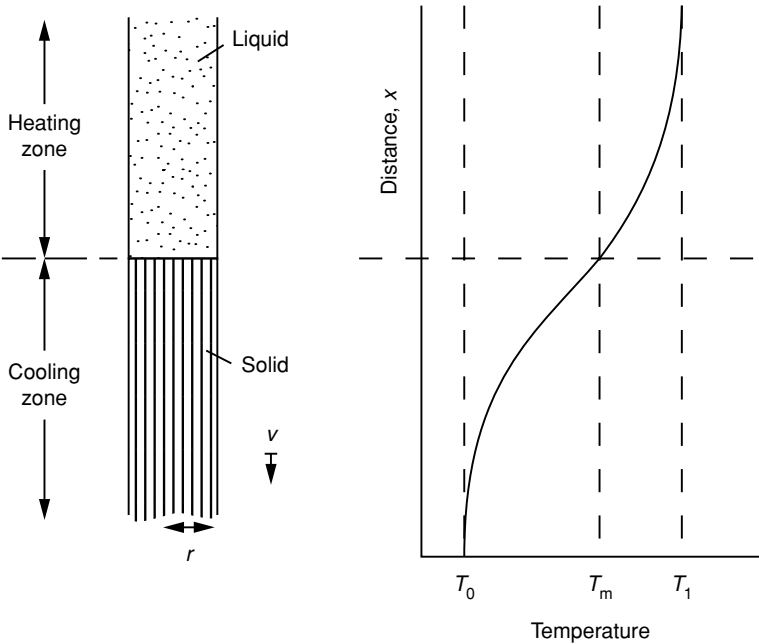


Fig. 3.16. The directional solidification of an infinite rod emerging from a semi-infinite furnace into a semi-infinite heat sink, at a velocity consistent with the temperature at the exit of the furnace equal to the melting temperature. Adapted from ref. [3].

heat transfer at the surfaces which have emerged from the furnace – the term h_{eff} is the associated heat-transfer coefficient. Under these circumstances, it can be shown that the temperature field is given by [10,11]

$$T\{x\} = T_0 + (T_m - T_0) \exp\{\beta_s x\}, \quad x < 0 \quad (3.3a)$$

and

$$T\{x\} = T_1 - (T_1 - T_m) \exp\{-\beta_l x\}, \quad x > 0 \quad (3.3b)$$

where

$$\beta_s = \left[\frac{2h_{\text{eff}}}{\kappa_s r} + \left(\frac{vc}{2\kappa_s} \right)^2 \right]^{1/2} - \frac{vc}{\kappa_s} \quad (3.4a)$$

and

$$\beta_l = \frac{vc}{\kappa_l} \quad (3.4b)$$

where the transition from the heating to the cooling medium is assumed to occur abruptly at $x = 0$; see Figure 3.16.

With these assumptions, one can derive an expression for the velocity at which the casting can be removed from the furnace, by substituting into the Stefan condition (given by Equation (3.2)) expressions for the temperature gradients at the solid/liquid interface – this is left as an exercise for the reader. One finds [11]

$$v = \frac{1}{c} \left(\frac{2h_{\text{eff}}\kappa_s}{r} \right)^{1/2} (Z^{1/2} - Z^{-1/2}) \quad (3.5a)$$

where

$$Z = \frac{L + c(T_1 - T_0)}{L + c(T_1 - T_m)} \quad (3.5b)$$

Note that if the casting speed is any greater than this value, solidification will not be completed before the casting leaves the furnace; this is not satisfactory since, in this case, a temperature gradient will be set up in the radial direction, which provokes the onset of defects such as freckles – this will be discussed further in Section 3.1.3. The analysis indicates that the critical velocity, v , is proportional to the square root of the heat transfer coefficient, h_{eff} . With $T_1 = 1800$ K, $T_0 = 300$ K, $T_m = 1650$ K, $L = 2 \times 10^9$ J/m³ and $c = 4 \times 10^6$ J/(m³ K) then $Z \sim 3$, so the last term in Equation (3.5a) is approximately unity. For a conventional investment casting furnace, a reasonable value for the heat transfer coefficient is $h_{\text{eff}} = 50$ W/(m² K), once a steady-state is attained. Taking $\kappa_s = 15$ W/(m K) and $r = 0.02$ m, one finds that the velocity, v , is about ~ 0.1 mm/s. This estimate is not very far from the values used in commercial furnaces for the casting of high-pressure turbine blades [9]. The temperature gradient, G , at the solidus can be determined from Equation (3.3a); it is $(T_m - T_0)\beta_s$, or about 75 K/cm, given the estimates that have been made.

In this analysis it has been assumed that there is no heat loss from the furnace to the cooling zone. This is impossible to achieve in practice, although much effort has been expended in designing optimum baffle arrangements by placing radiation shields at the base of the furnace, with openings just large enough to allow the castings to emerge from

the furnace. These have been shown to be efficient at keeping the solid/liquid interface flat, and in the vicinity of the baffle.

B Comparison of axial and transverse contributions to heat transfer

The above analysis indicates that, at steady-state, a balance exists between (i) the rate of heat extraction by conduction along the casting in the axial direction, and (ii) the rate of transverse heat loss by radiation through the curved, cylindrical surfaces. It follows that, as the casting first emerges from the furnace, this equilibrium has yet to be attained. The conditions pertaining to these early stages of directional solidification are important, particularly for single-crystal castings, since they govern the processes of competitive grain growth, texture development and grain selection.

At the beginning of directional solidification, the primary source of heat loss is due to conduction to the chill through the solidified portion of the ingot; this contribution diminishes as X , the length of the solidified ingot, increases. Thereafter, heat loss by radiation is expected to dominate, at least whilst the casting and ceramic investment shell remain in close contact and provided that the investment shell is relatively thin. The heat flux, q , should therefore be represented accurately by [3,12]

$$q = h_{\text{eff}}(T - T_0) = (h_C + h_R)(T - T_0) \quad (3.6)$$

where h_{eff} has now been partitioned into two terms, h_C and h_R , corresponding to the heat transfer coefficients for axial conduction and transverse radiation, respectively; these are given by

$$h_C = \frac{\kappa_s}{X} \quad (3.7)$$

and

$$h_R = \frac{\sigma(\epsilon T^4 - \alpha T_0^4)}{T - T_0} \quad (3.8)$$

where σ is the Stefan constant, ϵ is the emissivity of the ingot and α is the absorptivity of the surroundings.

It is helpful to determine the length of the ingot which solidifies before the steady-state is attained. If $\epsilon = \alpha = 0.45$ and $\sigma = 5.67 \times 10^{-8} \text{ J}/(\text{m}^2 \text{ s K}^4)$, $T = 1500 \text{ K}$ and $T_0 = 400 \text{ K}$, then

$$h_R = 120 \text{ W}/(\text{m}^2 \text{ K}) \quad (3.9)$$

and, with $\kappa_s = 16 \text{ W}/(\text{m K})$,

$$h_C = 1600 \text{ W}/(\text{m}^2 \text{ K}) \quad (3.10)$$

when $X = 10 \text{ mm}$. Thus, initially, $h_C > h_R$; however, at the point when $X = 130 \text{ mm}$ the two contributions are equal, and soon thereafter cooling is controlled by radiative heat transfer. The dominance of the radiative mode at later times is aided also by the relatively small cross-sectional area available for conduction through the copper chill, the presence of a grain selector and the substantial surface area of the mould, particularly for turbine blades which are of complex shape. It is clear therefore that, in practice, only the first part

of the casting is cooled effectively by conduction of heat through the water chill; for the majority of the casting one is relying upon radiation to the water-cooled walls of the vacuum chamber.

C Effects of quenching medium – liquid metal cooling

Equation (3.5a) indicates that the velocity of withdrawal, v , scales inversely with $\sqrt{r}$, the square root of the radius of the casting; thus, for castings of significant size, the time necessary for solidification to be completed can become prohibitively long. A further difficulty is that the economics of the single-crystal casting process depend also upon the yield, which is decreased if defects such as freckles, spurious grains and low-angle boundaries are formed; this is most likely to occur when the temperature gradient is low since the width of the semi-solid ‘mushy’ region is then considerable. The temperature gradient, G , is proportional to the parameter β_s in Equation (3.3a) – and Equation (3.4a) confirms that this decreases as r increases. Thus, the tendency for grain defect formation is enhanced for components of larger size. An additional challenge is that it becomes increasingly difficult to keep the profile of the solid/liquid interface flat as the casting becomes large; this can further increase the probability of defect formation, particularly freckles. These considerations are particularly pertinent to the casting of the blades for an industrial gas turbine (IGT) used for electricity generation – these might be about 500 mm long with a mass in excess of 5 kg.

Under these circumstances, more efficient methods are being sought for cooling the mould as it emerges from the furnace [13]. The rationale for this is as follows: a larger h_{eff} has the effect of increasing v and G , as indicated by Equations (3.3), (3.4) and (3.5). Enhanced cooling has been achieved in the laboratory by various means, including water sprays, fluidised bed cooling, gas cooling using jets of Ar/He mixtures, and liquid metal cooling; all depend upon a close thermal contact of the mould and the cooling medium. Of the different techniques, the liquid metal cooling (LMC) technique appears to be the most likely to make the transition into the commercial foundry – indeed it has been used already in Russia. The mould is immersed into a bath of molten metal to increase the heat transfer rate (see Figure 3.17). The choice of liquid metal coolant is important – the requirements are a low melting temperature, low vapour pressure, high thermal conductivity, low viscosity, lack of toxicity and low cost, on account of the quantities of metal required. For these reasons, aluminium and tin have emerged as the prime candidates for the LMC process. The important physical properties of the two metals are listed in Table 3.1. Although the vapour pressures of Al and Sn are more or less equivalent, the lower melting temperature of Sn is advantageous since it allows a lower bath temperature to be used. The thermal conductivity of Al is considerably higher than for Sn, which might indicate that heat transfer by conduction through the molten metal will be particularly effective.

In order to compare the advantages of Sn and Al as coolants for the LMC process, one must consider in more detail the various resistances to heat flow arising as the ingot and mould are immersed into the liquid metal. The conventional Bridgman process is considered first. The conduction and radiation coefficients, h_C and h_R , must be considered as before; the estimates of 1600 W/(m² K) and 120 W/(m² K) of Equations (3.9) and (3.10), respectively, are used again. However, for large IGT castings one must include the coefficient h_{gap} , to

Table 3.1. *The physical properties of two candidate metals for the liquid metal cooling (LMC) process – aluminium and tin [13]*

Property	Unit	Aluminium	Tin
Melting point	°C	660	232
Density	kg/m ³	2382	6980
Specific heat	kJ/(m ³ K)	2597	1857
Thermal conductivity	W/(m K)	104	33.5
Thermal diffusivity	m ² /s	40×10^{-6}	18×10^{-6}
Latent heat	kJ/m ³	922 000	419 000
Kinematic viscosity	m ² /s	6.3×10^{-7}	2.58×10^{-7}

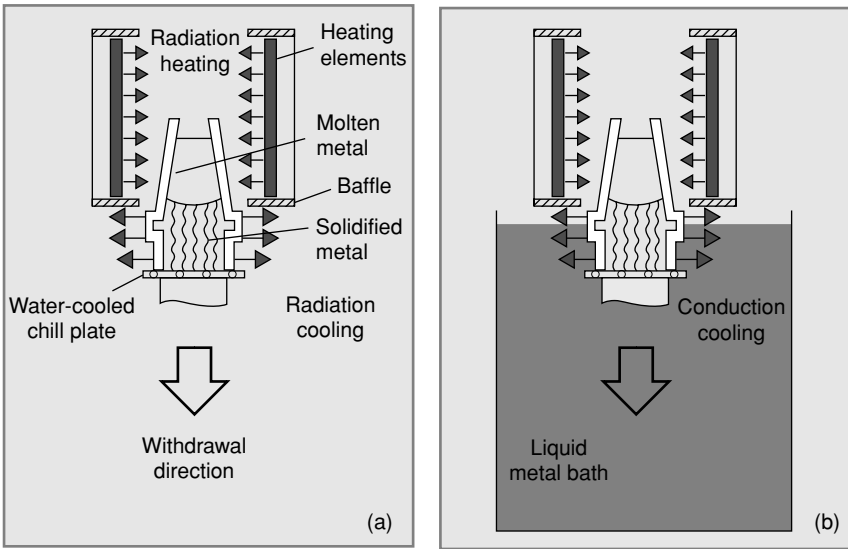


Fig. 3.17. Schematic illustration of two of the solidification processes used for casting. (a) Conventional high-rate-solidification (HRS) Bridgman process; (b) liquid metal cooling (LMC) process. (Courtesy of Robert Singer.)

acknowledge the need for radiation across the gap at the metal/shell interface, and h_{shell} for conduction through the investment shell, since it is relatively thick. A reasonable estimate for h_{gap} is $250 \text{ W}/(\text{m}^2 \text{ K})$. The conductivity of the porous ceramic shell is about $1.5 \text{ W}/(\text{m K})$, and a typical thickness for IGT applications is 12 mm ; this yields $h_{\text{shell}} = 125 \text{ W}/(\text{m}^2 \text{ K})$. The effective heat transfer coefficient, h_{eff} , is given by summing the four contributions, since the resistances to heat flow are in series; hence [14]

$$h_{\text{eff}}^{\text{Bridgman}} = \left[\frac{1}{h_C} + \frac{1}{h_{\text{gap}}} + \frac{1}{h_{\text{shell}}} + \frac{1}{h_R} \right]^{-1} \quad (3.11)$$

so that $h_{\text{eff}} = 50 \text{ W}/(\text{m}^2 \text{ K})$ for the conventional Bridgman process. With $T_1 = 1700 \text{ K}$ and $T_0 = 400 \text{ K}$, the net rate of heat transfer is $50 \times (1700 - 400)$ or $65 \text{ kW}/\text{m}^2$.

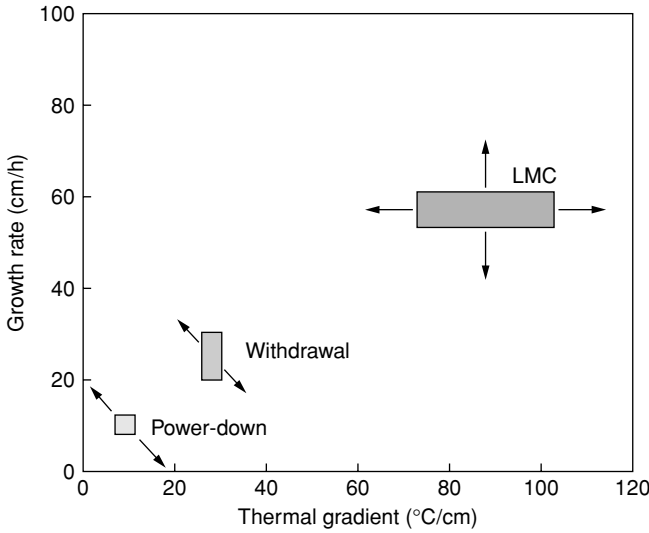


Fig. 3.18. Ranges of thermal gradients and growth rates for the three solidification processes appropriate for turbine blade castings: (i) the power-down process, (ii) the conventional withdrawal (Bridgman) process and (iii) the liquid metal cooling (LMC) process [12]. The arrows indicate directions in which changes in processing conditions can be accomplished easily.

For the LMC process, the contribution from h_R is no longer appropriate and it is replaced with a term, h_{LMC} , which describes the conduction of heat in the liquid metal. The liquid metal is an efficient heat sink so that $h_{LMC} \gg h_R$ – this is true regardless of whether Al or Sn is used as the coolant. Thus, h_{LMC} does not contribute significantly to the resistance to heat flow; hence,

$$h_{eff}^{LMC} = \left[\frac{1}{h_C} + \frac{1}{h_{gap}} + \frac{1}{h_{shell}} \right]^{-1} \quad (3.12)$$

so that $h_{eff} = 80 \text{ W}/(\text{m}^2 \text{ K})$. The net rate of heat transfer is then given by $h_{eff}(T - T_{bath})$, where T_{bath} is the bath temperature; appropriate values are 950 K and 550 K for Al and Sn, respectively. With $T = 1700 \text{ K}$, the net rates of heat transfer are then $80 \times (1700 - 950)$ or $60 \text{ kW}/\text{m}^2$ for Al and $80 \times (1700 - 550)$ or $90 \text{ kW}/\text{m}^2$ for Sn. These calculations confirm that the net rate of heat extraction can be improved markedly if molten Sn is used as a cooling medium. Liquid Al, on the other hand, does not improve matters appreciably – in fact, these calculations indicate that the net rate of heat transfer is lower than for the conventional Bridgman technique. This is a consequence of the higher bath temperature required by the higher melting temperature of Al.

In Figure 3.18, the growth rates and withdrawal velocities typical of the LMC process are compared with those which can be achieved using the conventional Bridgman (withdrawal) and power-down methods [12]. Liquid Sn is used as the cooling medium. One can see that the LMC process affords an increased thermal gradient of at least a factor of 3, and a growth velocity of at least a factor of 2, when compared to the conventional Bridgman method.

Fault	Six-month average	Fault	Six-month average
Orientation	8.0%	Inclusions	4.7%
High-angle boundaries	5.8%	High-angle boundaries	4.6%
Blocked core	3.1%	Core breakthrough	3.5%
Recrystallisation	3.0%	Root wall	3.4%
Inclusions	2.3%	Freckles	2.2%
Aerofoil shape	2.3%	Aerofoil shape	2.1%
Aerofoil wall	2.2%	Furnace malfunction	2.0%
Furnace malfunction	1.1%	Blocked core	1.4%
Overall yield	69%	Recrystallisation	1.1%
		Overall yield	67%

(a)

(b)

Fig. 3.19. Non-conformance statistics for two single-crystal castings processed in a commercial foundry: (a) seeded blade cast from second-generation alloy, and (b) blade cast from third-generation alloy, with a grain selector.

3.1.3 Formation of defects during directional solidification

A decision to cast a turbine blade aerofoil in single-crystal form represents a considerable challenge to the foundry. This is because very stringent requirements are then placed on the quality of the casting that can be accepted. Defects, such as high-angle grain boundaries, which arise when the grain selector fails to function properly or else due to nucleation of spurious grains around re-entrant features such as platforms and shrouds, cannot be tolerated. The crystallographic orientation must be such that the $\langle 001 \rangle$ direction is closely aligned to the axis of the casting – usually within 15° or so; in practice, the use of a grain selector or seed fails to ensure that this happens completely routinely. Recrystallisation during subsequent solutioning heat treatment, driven by plastic strains introduced by the shrinkage of the metal around the ceramic core, introduces high-angle grain boundaries, and therefore necessitates rejection [15]. Additional requirements, common to columnar or equiaxed castings used for lower-temperature applications such as low-pressure (LP) blading, must also be respected. The design of the component requires that tight dimensional tolerances are placed on the shape of the aerofoil and, if the blading is hollow, the thickness of its walls. The presence of entrained ceramic particles swept into the molten metal from the shell cannot be tolerated since they impair the fatigue properties; these are detected during the non-destructive inspection using radiographic methods. Given these considerations, it is clear that the probability of component rejection casting is likely to be reasonably high. The data in Figure 3.19, which are taken from a typical commercial foundry specialising in

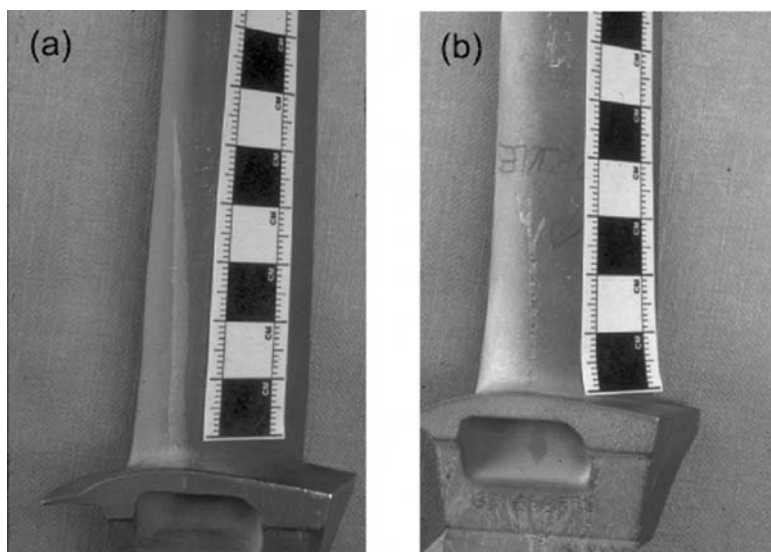


Fig. 3.20. Photographs of sections of blade castings illustrating the presence of defects: (a) high-angle grain boundaries; (b) freckle chains.

the casting of blading in single-crystal form, confirm this. For the seeded casting, failure to control the orientation of the blading is the major cause of non-conformance; this problem disappears when a grain selector is employed, but in both cases the overall yield is rather low at around 70%. Images of castings with grain defects are shown in Figure 3.20.

Case study: The freckle defect

The above data indicate that there are many possible defects which can arise when casting a turbine blade in single-crystal form. Of these, the phenomenon of freckling [16] is worthy of further consideration since the probability of its occurrence has been found to be strongly dependent upon the composition of the alloy being cast – the third- or fourth-generation alloys, which are rich in elements such as Re, are particularly prone to this effect. Freckles, also known as channel segregates, are thin chains of equiaxed grains associated with *localised* heterogeneities in chemical composition; see Figure 3.20(b). Single-crystal castings, both for aeroengine and IGT applications, can exhibit the freckling phenomenon. The *localisation* is always found to be on the scale of the primary dendrite arm spacing, with the freckles being found in the interdendritic regions. Chemical analysis of the freckle chains confirms that they are enriched with regard to elements which partition to the interdendritic liquid, for example Al, Ti, Ta and Nb, where these are present.

There are other pieces of evidence which shed light on the mechanism by which freckles form. Consider Figure 3.21, which summarises the results from a considerable programme of experimental work carried out by Tin, Pollock and co-workers aimed at identifying the factors which promote freckling [17]. More than 16 different alloys were cast into cylindrical

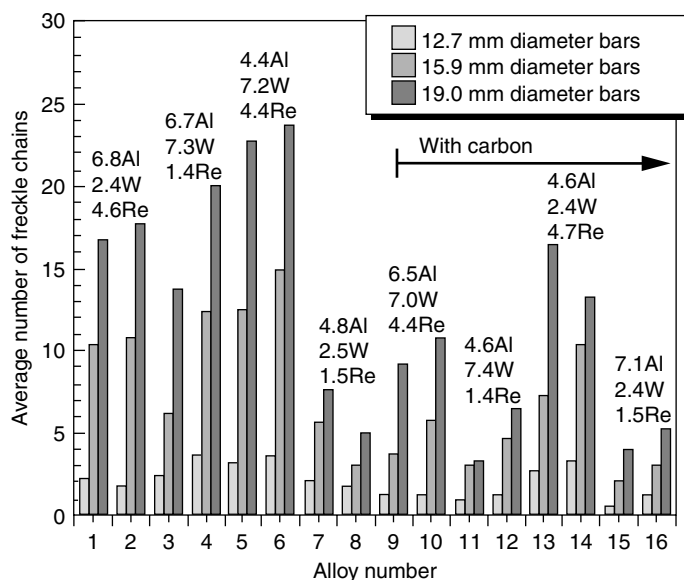


Fig. 3.21. Statistics for the formation of freckle chains in cast bars of varying diameters, in superalloys of various compositions [17]. For all alloys, Ta $\sim$ 6.1, Hf $\sim$ 0.14. All numbers refer to concentrations in wt%.

bars of three different diameters: 12.7 mm, 15.9 mm and 19.0 mm, at otherwise identical conditions. Some of the alloys were deliberately doped with carbon. The average number of freckle chains appearing on the surface of the castings was then characterised. A number of interesting observations are seen. First, the extent of freckling becomes more pronounced as the bar diameter increases. Second, there is a greater tendency for freckling when the combined W and Re content, i.e. W + Re, is significant. Third, additions of carbon are found to be effective at reducing the degree of freckling. Finally, and significantly, when casting is carried out at different casting conditions (see Figure 3.22) it is found that there is a critical minimum primary dendrite arm spacing (corresponding to a lower limit for the temperature gradient) below which freckles are not found [18].

These observations support the following mechanism for the freckling effect. On thermodynamic grounds, the alloying additions prefer to partition to either the interdendritic regions or alternatively to the dendrites themselves. Experiments using electron probe microanalysis (EPMA) indicate that Re and W segregate to the dendrite core, and that Al, Ti and Ta partition to the interdendritic regions [19]; see Figure 3.23. Other elements such as Cr, Co and Ru are found to be rather neutral in partitioning behaviour, with a partitioning coefficient close to unity. Particularly when the Re and W contents are large, liquid in the interdendritic region becomes depleted in these heavy elements, causing it to become lighter than the liquid at a temperature closer to the liquidus. Thus, a so-called ‘density inversion’ becomes possible [20]. For freckling to occur, the enhanced buoyancy of the interdendritic liquid (due to the loss of heavy elements) must overcome frictional effects which prevent the lighter liquid from rising – if this occurs, then thermosolutal convection currents, or

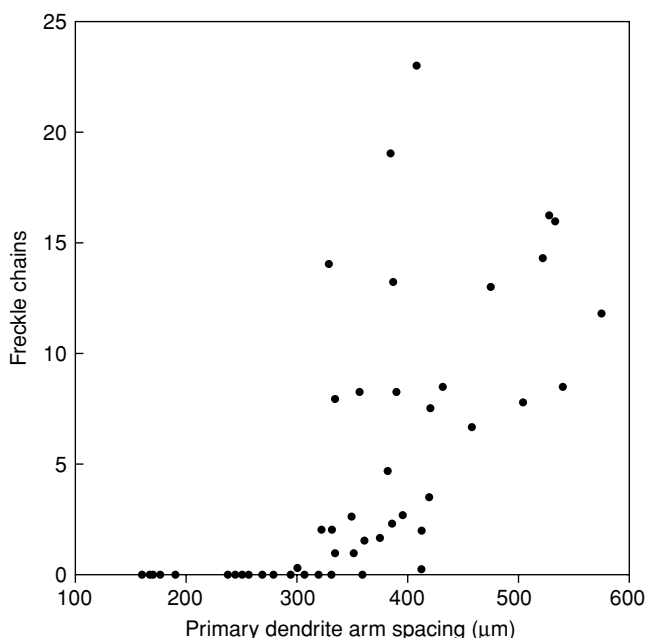


Fig. 3.22. Dependence of the number of freckle chains on the primary dendrite arm spacing. Note that the critical spacing for the formation of freckles is $\sim 300 \mu\text{m}$ [18].

‘freckle plumes’, are set up. When casting is of significant cross-sectional area, the width of the mushy zone is wide, and freckling is exacerbated. Similarly, additions of elements such as Re and W, which exacerbate the density inversion on account of their partitioning behaviour, promote freckling; conversely, elements such as Ta restrict it. Additions of carbon cause the formation of carbides at temperatures around the liquidus, and these appear to be responsible for its beneficial effect [17]. When the dendrite arm spacing is low, freckling does not occur since any possible density inversion provides insufficient driving force to overcome the resistance to interdendritic fluid flow, since the permeability of the mushy zone is then low.

That the single-crystal superalloys are prone to the freckle defect can be seen by making calculations for the density of the liquid phase as the function of temperature; see Figure 3.24. Experimental measurements for the density of liquid CMSX-4 above the melting temperature (about 1650 K) have been reported in ref. [21]; these data are plotted on the figure, along with an extrapolation to lower temperatures (the dotted line), which represents the density of the *unpartitioned* liquid in the mushy zone. Estimates of the density of the *partitioned* liquid can be made assuming equilibrium is attained, by accounting for the mass loss due to the partitioning of elements such as Re and W into the solid phase. The density of the partitioned liquid is given by the curved solid line. The driving force for freckling is proportional to the *difference* between the solid and dotted lines, denoted $\Delta\rho$; for CMSX-4 it is a maximum at about 20 K below the melting temperature. These calculations can be carried out for single-crystal superalloys of different compositions, for example